

Vitamin	AKA	Coenzyme Form	Biochemical Function	Deficiency/Associated Diseases	Causes	Extras
A	Retinol		Vision: All-<i>trans</i> retinol → 11-<i>cis</i> retinol → 11-<i>cis</i> retinal → rhodopsin → all-<i>trans</i> retinal Epithelium: Retinol → retinoic acid (cytosol) → retinoic acid-receptor complex (nucleus) binds to chromatin → transcription of keratin	Deficiency: night blindness, xerophthalmia, bitot spots, keratomalacia, increased risk of pulmonary infections, immune deficiency Hypervitaminosis A: skin peeling, headaches, benign intracranial HTN, enlarged liver; teratogen in pregnancy	Fat-free diet, fat malabsorption (Cystic Fibrosis, obstructive jaundice)	Therapeutic uses: Retinoic Acid treats acne, measles, & acute promyelocytic leukemia β-carotene (provitamin A) – Antioxidant
B1	Thiamine	TPP (thiamine pyrophosphate)	PPP: Transketolase PDH: Pyruvate D.H. Kreb: α-KG D.H. Branched Chain AA Synthesis: Branched Chain Keto Acid D.H.*	Beriberi: B1 deficiency causing polyneuropathy progressing to paralysis (dry), cardiovascular failure (wet) Wernicke-Korsakoff: associated with EtOH abuse, results in confusion, ophthalmoplegia, ataxia, nystagmus, memory loss, confabulation *MSUD	Diet of polished rice, white flour, & white sugar	Diagnostic test: ↓ erythrocyte transketolase activity
B2	Riboflavin	FMN (flavin mononucleotide), FAD (flavin dinucleotide)	PDH: Dihydrolipoyl D.H. Kreb: Succinate D.H. ETC: Glycerol 3-P D.H. Lipolysis: Acyl CoA D.H.*	Deficiency: cheilosis, glossitis, facial dermatitis *MCAD, Jamaican Vomiting Sickness		
B3	Niacin	NAD ⁺ , NADP ⁺	NAD⁺ Glycolysis: Glyceraldehyde 3-P D.H. Cori: Lactate D.H. PDH: Dihydrolipoyl D.H. Kreb: Isocitrate D.H., α-KG D.H., Malate D.H. Lipolysis: 3-hydroxyacyl CoA D.H. (2nd round oxidation) Ketogenesis: β-hydroxybutyrate D.H. Ethanol: Alcohol D.H., Acetaldehyde D.H. NADP⁺	Deficiency: pellagra (Dermatitis, Diarrhea, Dementia, Death) Hartnup Disease: AR; deficiency of tryptophan transporters in proximal renal tubular cells & enterocytes Carcinoid Syndrome: excessive conversion of tryptophan to serotonin	Corn based diets due to deficiency in niacin & tryptophan (precursor)	Therapeutic uses: Niacin inhibits lipolysis in adipose tissue; used to treat Type IIb Hyperlipoproteinemia

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B5			PPP: Glucose 6-P D.H., 6-Phosphogluconate D.H. FA Synthesis: Malic enzyme Urea: Glutamate D.H.			
	CoA	Pantothenic acid				
B6	Pyridoxine	PLP (pyridoxal phosphate)	Glycogenolysis: Glycogen Phosphorylase* AA Metabolism: ALT, AST Heme Synthesis: ALA Synthase NT Synthesis: DOPA D.H., AA D.H. Homocysteine: Cystathionine β -Synthase	Deficiency: microcytic anemia, peripheral neuropathy, increased risk of CVD, seizures *McArdle Disease, Hers Disease	Isoniazid (anti-TB drug) inactivates Pyridoxine	
B7	Biotin		Gluconeogenesis: Pyruvate carboxylate FA Synthesis: ACCase Odd-Chain FA Metabolism: Propionyl-CoA carboxylase	Multiple Carboxylase Deficiency	Avidin in raw egg whites can inhibit absorption of Biotin	
B9	Folic Acid	Methyl-THF	1-C transfer/methylation leading to synthesis of nitrogenous bases in DNA & RNA Step in homocysteine pathway	Deficiency: macrocytic, megaloblastic anemia	Low vegetable diet, impaired absorption in jejunum, increased requirement (pregnancy), use of Methotrexate (folate antagonist), trapping due to B12 deficiency	
B12	Cobalamin		Odd-Chain FA Metabolism: Methylmalonyl CoA Mutase Homocysteine: Methionine synthase	Deficiency: neurological manifestations (from accumulation of methylmalonate, interferes with myelin formation), macrocytic megaloblastic anemia	Damage to distal ileum (req. intrinsic factor from gastric parietal cells) or parietal cells (pernicious anemia), diet low in animal sources (veganism)	Causes FOLATE TRAP
C	Ascorbic Acid		NT Synthesis: Dopamine β -hydroxylase Collagen Synthesis: post-trans mod. of hydroxyproline and hydroxylysine residues, creates hydrogen bonds	Scurvy: bleeding gums, loose teeth, perifollicular hemorrhages, bleeding into joints, ecchymosis, impaired/delayed wound healing, microcytic anemia (due to reduced iron absorption)	Diet low in fruits and vegetables	Therapeutic uses: slows progression of age-related macular degeneration Antioxidant
D	Calciferol		(Skin) 7-dehydrocholesterol $\rightarrow$ cholecalciferol $\rightarrow$ (liver) 25-hydroxycholecalciferol $\rightarrow$ (kidney) 1,25-dihydroxycholecalciferol/ calcitriol $\rightarrow$	Rickets: soft pliable bones, bow-legged & pigeon chest deformity, rachitic rosary, delayed closure of fontanelle, delayed teething	Reduced exposure to sunlight, renal disease, liver disease, fat malabsorption	Diagnostic tests: $\uparrow$ ALP, $\downarrow$ serum Ca^{2+} , $\downarrow$ serum phosphate

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Minerals			1,24,25-dihydroxycholecalciferol (INACTIVE) Calcitriol (1) increases calbindin in intestines to increase absorption of Ca^{2+} & phosphate, (2) stimulates mobilization of Ca^{2+} & phosphate from bone in presence of PTH, and (3) inhibits Ca^{2+} excretion in kidney	Osteomalacia: adulthood onset, non-specific bone pain, increased risk of fractures Calcitriol Receptor Mutation: low serum Ca^{2+} & phosphate, elevated 1,25-hydroxyvitamin D Hypervitaminosis D: hypercalcemia, hyperphosphatemia, ectopic soft tissue mineralization		
	E	Tocopherols	Prevents peroxidation of membrane lipids, prevents formation of oxidized VLDL, scavenges free radicals	Deficiency: hemolytic anemia (due to abnormal cell membranes), reduced deep tendon reflexes & gait problems (due to axonal degeneration)	Fat malabsorption	Therapeutic uses: slows progression of age-related macular degeneration Antioxidant
	K	Phylloquinones (plants)/ Menaquinones (bacteria)	Synthesis of clotting factors II, VII, IX, & X: γ-Glutamyl carboxylase Synthesis of Proteins C & S (anticoagulant system)	Deficiency: hematuria, melena, ecchymosis, bleeding gums Hemorrhagic Disease of Newborn: bleeding from skin, umbilicus, viscera, & intracranially; due to sterility of intestines → no synthesis of Vitamin K	Diet low in vegetables, fat malabsorption, prolonged use of broad-spectrum antibiotics	Diagnostic tests: ↑ INR/Prothrombin time Warfarin (anticoag drug) is a Vitamin K antagonist; blocks activity of VKOR to prevent recycling
	Minerals		Copper ETC: Cytochrome oxidase (complex IV) Collagen Synthesis: Lysyl oxidase, creates covalent cross-linkages Melanin Synthesis: Tyrosinase	Menkes Syndrome: XR; defective absorption from GIT → gray, “kinky” hair, aneurysm, cerebral dysfunction, microcytic anemia Wilson Disease: AR, allelic heterogeneity, late onset; mutation of copper transporting ATPase → accumulation of copper in liver, brain, & eyes → hepatitis, cirrhosis, Kayser Fleischer rings, neuropsychiatric symptoms, hemolysis *Lab results: ↓ serum ceruloplasmin, ↑ urinary excretion of copper & hepatic copper content		Forms ceruloplasmin (ferroxidase): - transports copper - converts ferrous iron (absorbed) → ferric & incorporates it into transferrin for transport - mobilizes ferric ions from ferritin & hemosiderin
		Iron	ETC: Cytochrome reductase (complex III), cytochrome c, cytochrome oxidase (complex IV) Heme Synthesis: incorporated into heme of hemoglobin & myoglobin	Iron Deficiency Anemia: most common nutritional deficiency; hypochromic microcytic anemia, fatigue, pallor, weakness, dizziness, brittle nails, pica *Lab results: ↓ serum ferritin & iron, <20% saturation of transferrin, ↓ H&H, ↓ MVC & MCHC, ↓ bone marrow iron stores, ↑ TIBC Hemochromatosis: AR, allelic heterogeneity, late onset; mutation of HFE gene → excessive absorption of iron → joint pain, fatigue, hepatomegaly, DM, cardiac dysfunction, brownish skin pigmentation *Lab results: ↑ serum ferritin & iron, >50% saturation of transferrin, ↓ TIBC	Anemia: veganism, low dietary vitamin C, achlorhydria, high dietary fiber, pregnancy, chronic bleeding	Serum iron (ferric) always bound to transferrin Stored in liver and RES as ferritin & hemosiderin Hemochromatosis has variable expressivity: M present earlier than F because menstruation is protective